

Mathematics(P&C)- A Das Gupta

Time Allowed : 1 hour _____ **Maximum Marks : 35**

Please read the instructions carefully. You will be allotted 5 minutes specifically for this purpose.

Instructions

A. General

1. Blank papers, clipboards, log tables, slide rules, calculators, cellular phones, pagers, and electronic gadgets in any form are not allowed.
2. Do not break the seals of the question-paper booklet before instructed to do so by the invigilators.

B. Question paper format and Marking Scheme :

1. Section I contains 5 questions. Each question in this section contains 3 marks. 1 mark will be deducted for a wrong response.
2. Section II contains 1 question. The question carries 5 marks. 2 marks will be deducted for a wrong response.
3. Section II contains 3 questions based on a comprehension. Each question carries 5 marks. 2 marks will be deducted for a wrong response.

SECTION – I

This section contains 5 questions. Only one option is correct.

Q1: If $\begin{vmatrix} x & 4 & 7 & 9 \\ 0 & 1 & 5 & 8 \\ 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 3 \end{vmatrix} =$ the total number of arrangements of 4 different things, taking at least one at a time, then x is equal to

- A) $\frac{11}{8}$
- B) 4
- C) $\frac{11}{2}$
- D) $\frac{32}{3}$

Q2: Let $C_1, C_2, \dots, C_n, \dots$ be a sequence of concentric circles. The n th circle has the radius n and it has n openings. A point P starts travelling on the smallest circle C_1 and leaves it at an opening along the normal at the point of opening to reach the next circle C_2 . Then it moves on the second circle C_2 and leaves it likewise to reach the third circle C_3 and so on. The total number of different paths in which the point can come out of the n th circle is

- A) $2^n \cdot n!$
- B) $2^{n-1} \cdot n!$
- C) $n!$
- D) $2^{n-1} \cdot (n-1)!$

Q3: The number of ways of selecting two numbers from the first 15 natural numbers such that their sum is a multiple of 5, is

- A) 20
- B) 36
- C) 21
- D) 16

Q4: The number of ways, in which 9 flowers, of which 5 are identical and white, and the other four are of different colours, can be set on a garland so that no two of the coloured flowers are consecutive, is

- A) 60
- B) 120
- C) 180
- D) None of these

Q5: ${}^nC_{r+1} + {}^nC_{r-1} + 2 \times {}^nC_r$ is equal to

- A) ${}^{n+2}C_{r+1}$
- B) ${}^{n+1}C_r$
- C) ${}^{n+1}C_{r+1}$
- D) ${}^{n+2}C_r$

SECTION – II

This section contains 1 question. More than one options are correct

Q6: There are $n + 1$ identical balls on which the numbers ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$ are written, each ball carrying a different number. The total number of ways in which the balls can be arranged in a row so that the values of numbers on the first and the last are equal, is

- A) $n!$ when n is even
- B) $(n^2 - 1) \cdot (n - 2)!$ when n is odd
- C) $\frac{1}{2} \cdot n!$ when n is even
- D) $\frac{1}{2} (n + 1) \cdot (n - 1)!$ when n is odd

SECTION – III

This section contains 1 comprehension .

Comprehension 1: $2^m 3^n 5^p$ is a divisor of $2^{10} \cdot 3^8 \cdot 5^7$ if m, n and p are all whole numbers such that $0 \leq m \leq 10, 0 \leq n \leq 8$ and $0 \leq p \leq 7$. Also, $(2^0 + 2^1 + 2^2 + \dots + 2^9) (3^0 + 3^1 + 3^2 + \dots + 3^7) (5^0 + 5^1 + 5^2 + \dots + 5^6)$ = sum of all the divisors of $2^{10} \cdot 3^8 \cdot 5^7$.

Q7: The number of proper divisors of 16200 is

- A) 60
- B) 24
- C) 58
- D) 22

Q8: The sum of proper divisors of 360 is

- A) 1170
- B) 810
- C) 1169
- D) 809

Q9: The sum of odd divisors ($\neq 1$) of 10800 is

- A) 1239
- B) 1240
- C) 1238
- D) 2479